

Quantum Machine Learning for Early Disease Prediction

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Abstract— Early disease prediction is an important application of artificial intelligence in healthcare, where timely identification of disease risk can support clinical decision-making and improve patient management. Quantum Machine Learning (QML) has emerged as a promising research direction by combining quantum computing principles with machine learning techniques for healthcare prediction tasks. This paper reviews recent research on QML-based disease prediction, with emphasis on quantum algorithms, healthcare datasets, reported outcomes, and practical limitations. The reviewed studies demonstrate the use of approaches such as Quantum Support Vector Machines (QSVM), Quantum Neural Networks (QNN), Variational Quantum Circuits (VQC), quantum ensemble methods, and optimization-based techniques for disease prediction. In addition to the review, a QML-based experimental framework is implemented using the UCI Heart Disease dataset to examine the application of quantum kernel methods for disease prediction. The implementation includes data preprocessing, feature scaling, Principal Component Analysis (PCA), quantum feature mapping, Fidelity Quantum Kernel, and Quantum Support Vector Classification (QSVC), with a classical Support Vector Classifier (SVC) used for comparison. The analysis also highlights practical challenges such as limited dataset size, dependence on quantum simulators, feature reduction, and restricted quantum hardware resources. The study provides a structured basis for further research toward scalable, robust, and clinically applicable QML-based healthcare systems.

Index Terms— Quantum Machine Learning, Quantum Computing, Early Disease Prediction, Heart Disease, Healthcare, QSVM, QSVC, Fidelity Quantum Kernel, PCA, Clinical Decision Support.

I. INTRODUCTION

Early disease prediction is increasingly important in healthcare because recognizing disease-related risk before severe symptoms develop can enable earlier clinical attention. Modern healthcare systems collect information through electronic health records, medical images, wearable devices, and genomic resources. These sources provide valuable signals but also create data-processing challenges because the information can be heterogeneous, high-dimensional, and difficult to analyze consistently. Machine learning has become useful for diagnosis, risk estimation, image analysis, and decision support, while conventional models continue to face challenges when healthcare data becomes complex and difficult to process.

Quantum computing provides an alternative computational paradigm based on quantum states and operations. Quantum Machine Learning combines these ideas with machine-learning workflows and has been explored for classification, feature representation, optimization, and prediction. Recent healthcare studies have investigated models including QSVM, QNN, and VQC for disease-related tasks, but many experiments remain limited to simulators, small datasets, and controlled settings. These conditions make it difficult to determine whether reported improvements will translate into reliable clinical systems.

This paper reviews recent work on QML for early disease prediction and identifies recurring technical and deployment gaps. In addition to the review, a QML-based experimental framework is implemented using the UCI Heart Disease dataset to examine quantum-kernel-based disease prediction. The implementation applies data preprocessing, feature scaling, Principal Component Analysis (PCA), quantum feature mapping, Fidelity Quantum Kernel, and Quantum Support Vector Classification (QSVC), with a classical Support Vector Classifier (SVC) used as a comparison model. Rather than claiming a demonstrated clinical quantum advantage, the study evaluates the practical application of QML under a controlled experimental setting and discusses its limitations and future validation needs.

II. LITERATURE REVIEW

A. QML in Healthcare

Recent literature describes QML as an emerging intersection of quantum computing and machine learning for healthcare analytics. Review studies report applications across medical imaging, disease prediction, personalized medicine, genomics, and decision support. At the same time,

the literature repeatedly notes that current quantum devices are constrained by noise, limited qubit counts, and imperfect gates, which restrict the scale of healthcare experiments. Digital-health reviews also emphasize that evidence from real clinical deployments remains limited and that comparisons with strong classical baselines are not yet uniform [1], [3], [8].

B. QML for Disease Prediction

Disease prediction is one of the important application areas for QML because classification models can operate on clinical features or transformed representations of medical data. Cardiovascular studies have examined quantum support-vector approaches, Pegasos QSVC, and quantum neural architectures [2], [5], [6]. In the reviewed cardiovascular work, a QNN-based model reported 97.31% accuracy on a multi-class ECG-image task, while quantum classifiers also showed improvements over selected classical configurations [5]. These findings are encouraging but should be interpreted within the experimental settings, dataset characteristics, and simulator-based execution used in the studies. The existing literature provides a useful basis for investigating quantum-kernel-based approaches for heart disease prediction using structured clinical features.

C. Current Challenges

Across the reviewed literature, four limitations appear repeatedly. First, many experiments use quantum simulators instead of physical quantum processors. Second, datasets are often small, specialized, or disease-specific, which limits generalization. Third, data encoding and feature reduction can become bottlenecks because classical healthcare data must be represented in a form suitable for quantum circuits. Finally, integration into hospital workflows requires attention to interoperability, patient privacy, regulatory requirements, explainability, and clinical validation [7], [8].

D. Research Trends

Current research increasingly favors hybrid quantum-classical workflows that use classical processing for data preparation and optimization while reserving quantum circuits for selected computational steps. Other directions include improved feature maps, variational circuits, quantum kernels, noise mitigation, and benchmarking against classical methods. Future work is moving toward larger datasets and more realistic hardware experiments, although practical quantum advantage in healthcare has not yet been established consistently [7], [8].

III. COMPARATIVE ANALYSIS OF REVIEWED STUDIES

TABLE I. COMPARISON OF RECENT QML STUDIES FOR EARLY DISEASE PREDICTION

Ref.	Disease / Application	QML Algorithm	Key Finding	Limitation	Future Work
[1]	Healthcare	Multiple QML approaches	Reviews emerging QML applications across healthcare and identifies potential areas for future research	Limited real-world clinical validation and hardware maturity	Evaluate QML on larger datasets and real quantum hardware
[2]	Heart disease	QML + QPSO	QML with preprocessing and QPSO achieved 96.7% accuracy, outperforming the evaluated classical MLP model	Dataset and computational setting limit generalization	Evaluate with broader datasets and realistic quantum environments
[3]	Medical decision support	QC + ML / QML approaches	Reviews how quantum computing and ML can support diagnosis, prediction, optimization, and clinical decision-making	Many applications remain experimental or conceptual	Develop real-world case studies and clinically validated implementations
[4]	Disease detection and prediction	Various QML approaches	Surveys QML applications for disease detection and prediction across multiple healthcare domains	Limited hardware maturity and practical validation	Develop benchmarks, improved feature encoding, and hardware-based evaluation
[5]	Cardiovascular disease	QSVC, Pegasos QSVC, QNN	QNN achieved 97.31% accuracy and outperformed the classical CNN; QSVC also outperformed SVC.	Evaluation depends on the selected dataset and experimental setup	Test on broader cardiovascular datasets and quantum hardware
[6]	Heart disease	Quantum Ensemble Learning (QuEML)	QuEML achieved a 0.6% accuracy improvement and reduced average training time compared with the evaluated classical approaches	Improvement was modest and evaluation used a Kaggle dataset	Validate on larger clinical datasets and optimize the ensemble
[7]	QML theory and applications	QNN, QBoost and related methods	Identifies important QML challenges including trainability, noise, scalability, and achieving quantum advantage	Noise, scalability, and hardware limitations remain unresolved	Develop error mitigation, fault-tolerant methods, and improved algorithms
[8]	Digital health	Various QML models	Systematic review finds growing QML research in digital health but limited consistent evidence of practical quantum advantage	Few realistic hardware evaluations and inconsistent benchmarking	Improve hardware-based evaluation, robustness, scalability, and standardized benchmarking

IV. RESEARCH GAP

The literature indicates that QML is promising for selected disease-prediction tasks, but the evidence base is not yet sufficient for routine healthcare deployment. A major gap is the difference between algorithm-level demonstrations and practical end-to-end workflows. Many studies evaluate a classifier on a particular disease or dataset, while a practical healthcare system must also consider data preprocessing, feature representation, prediction, interpretation, and integration with clinical processes.

A second gap concerns validation and comparison. Simulator-based experiments are useful for developing and evaluating quantum models, but they do not capture all effects of physical hardware noise, calibration drift, connectivity constraints, and execution overhead. Small or specialized datasets also make it difficult to determine whether reported results can generalize to other patient populations. In addition, standardized evaluation procedures for comparing QML models with classical baselines remain inconsistent across studies.

To address these gaps, this study investigates a quantum-kernel-based approach for heart disease prediction using the UCI Heart Disease dataset. The experimental workflow combines classical preprocessing and PCA-based feature reduction with a quantum feature map, Fidelity Quantum Kernel, and QSVC. A classical SVC is evaluated using the same matched training subset and test set to provide a controlled comparison. This experimental setup provides a practical basis for examining the applicability and

limitations of QML for heart disease prediction without claiming a demonstrated quantum advantage.

V. PROPOSED FRAMEWORK

The proposed framework follows an end-to-end workflow for heart disease prediction using a combination of classical preprocessing and quantum machine learning techniques. The UCI Heart Disease dataset is first collected and prepared for analysis. Preprocessing handles missing values and converts the clinical attributes into a suitable numerical representation. The features are standardized and Principal Component Analysis (PCA) is applied to reduce the original feature space to four components for quantum processing.

The reduced features are then mapped using a quantum feature map based on the ZZFeatureMap. A Fidelity Quantum Kernel is used to measure the similarity between quantum feature representations, and the resulting kernel is provided to a Quantum Support Vector Classifier (QSVC) for heart disease classification. A classical Support Vector Classifier (SVC) is also evaluated using the same matched training subset and test set to provide a controlled comparison.

The prediction stage produces a classification indicating the presence or absence of heart disease. The resulting performance is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix. The system is intended as an experimental decision-support framework rather than an autonomous clinical diagnosis system.

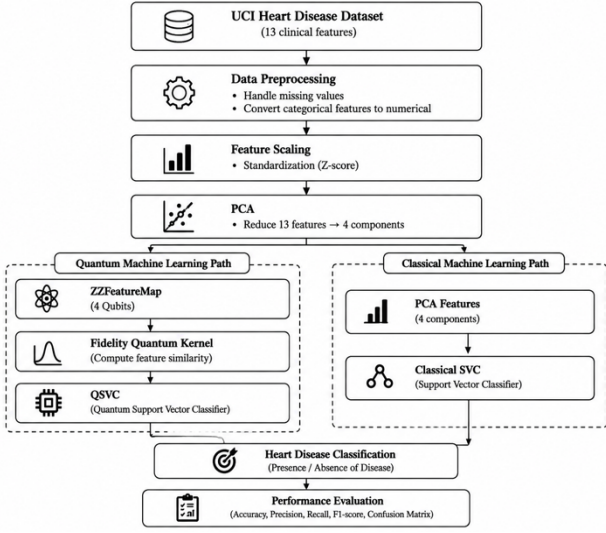


Fig. 1. Proposed conceptual framework for quantum machine learning-based early disease prediction.

VI. IMPLEMENTATION AND EXPERIMENTAL RESULTS

The proposed framework was implemented as a proof-of-concept binary disease-classification pipeline using the **UCI Heart Disease dataset**. The dataset contains **303 samples with 13 clinical features** and a binary target indicating the presence or absence of heart disease. The implementation used Python, scikit-learn for preprocessing and the classical baseline, and Qiskit Machine Learning for the quantum-kernel model. The workflow used an **80:20 stratified train/test split**, resulting in **242 training samples and 61 test samples**. The clinical features were standardized and Principal Component Analysis (PCA) was applied to reduce the 13-dimensional feature space to four components for quantum processing.

Fig. 2 shows the implemented system architecture, including the classical SVC baseline and the quantum-kernel classification stage.

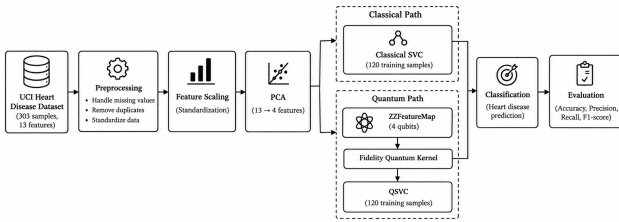


Fig. 2. Implemented system architecture for the QML-based heart disease prediction pipeline.

The implementation flow is summarized in **Fig. 3**. To provide a controlled comparison, a **stratified subset of 120 training samples** was selected after PCA and used for **both the classical SVC and QSVC models**. The same **61-sample test set** was retained for evaluating both models. The quantum model uses a four-qubit ZZFeatureMap with a Fidelity Quantum Kernel, followed by QSVC classification. The matched training configuration was used because fidelity quantum-kernel simulation is computationally expensive.

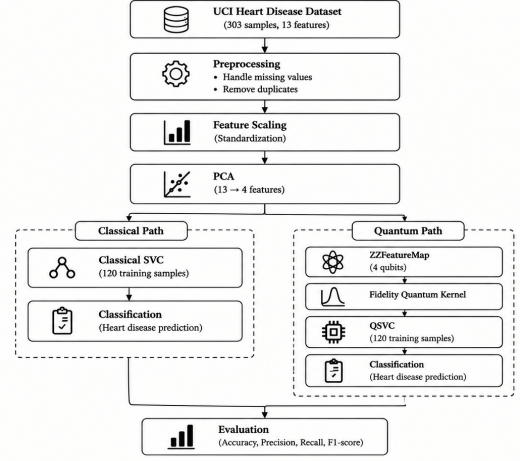


Fig. 3. Flowchart of the implemented experimental workflow.

The results in Table II show that the classical SVC achieved higher performance across all evaluation metrics compared with QSVC. The QSVC obtained an accuracy of 44.26%, while the classical SVC achieved 88.52%. The lower QSVC performance can be attributed to the limited training subset, PCA-based feature reduction, quantum feature-map configuration, and the computational constraints of quantum-kernel simulation. Therefore, the results demonstrate the feasibility of implementing a quantum-kernel-based approach rather than claiming a quantum advantage.

TABLE II. PERFORMANCE COMPARISON OF CLASSICAL SVC AND QSVC

Metric	Classical SVC	QSVC
Accuracy	88.52%	44.26%
Precision	86.21%	31.25%
Recall	89.29%	17.86%
F1-score	87.72%	22.73%

The QSVC confusion matrix in Fig. 4 presents the classification results obtained on the 61-sample test set. The matrix provides the numbers of correctly and incorrectly classified instances for the two heart disease classes. The corresponding accuracy, precision, recall, and F1-score are reported in Table II. The results are interpreted within the controlled experimental setting, considering the use of PCA-based feature reduction, a limited matched training subset, and quantum-kernel simulation.

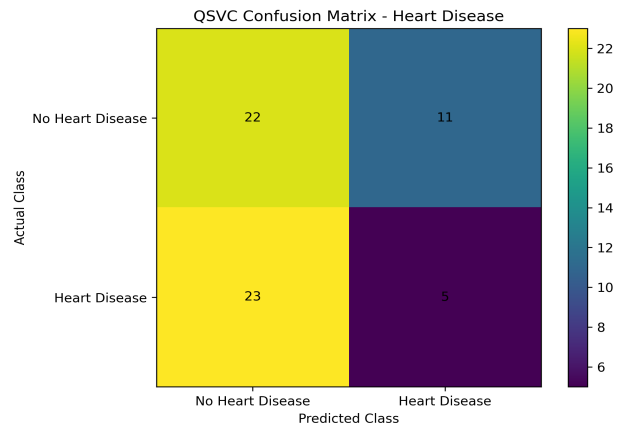


Fig. 4. Confusion matrix of the QSVC model on the test dataset.

Fig. 5 compares the four evaluation metrics for the classical SVC and QSVC models. The experiment demonstrates an end-to-end QML workflow for heart disease classification and provides a controlled comparison between classical and quantum-kernel approaches using the same training and test samples. The results also highlight the practical effects of feature reduction, limited training data, and quantum-kernel simulation on model performance.

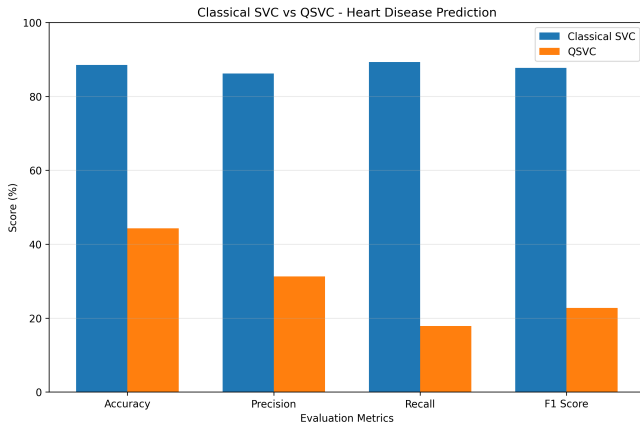


Fig. 5. Performance comparison of Classical SVC and QSVC.

VII. DISCUSSION

The experimental results demonstrate that QML can be applied to structured heart disease data using a quantum-kernel-based classification approach. However, the results also show that QML performance should not be assumed to be superior to classical machine learning. The observed performance depends on factors such as dataset size, feature reduction, quantum feature-map design, kernel configuration, training-set size, and the use of quantum simulation. Therefore, the results should be interpreted as an experimental evaluation of QML rather than evidence of universal quantum superiority.

The proposed implementation places quantum processing within a complete data-to-prediction workflow consisting of preprocessing, feature scaling, PCA-based dimensionality reduction, quantum feature mapping, Fidelity Quantum Kernel computation, and QSVC classification. The use of the same matched training subset and test set for SVC and QSVC provides a more controlled basis for comparison. At the same time, the limited dataset size and computational cost of quantum-kernel simulation restrict the scale of the present experiment.

These observations indicate that future QML healthcare studies should evaluate not only classification performance but also data quality, feature representation, computational requirements, hardware constraints, explainability, privacy, and clinical validation. The framework therefore provides a practical basis for further investigation of quantum machine learning for early disease prediction.

VIII. LIMITATIONS AND FUTURE WORK

The implemented system is a proof-of-concept experimental model and has not been presented as a clinically validated diagnostic system. The evaluation is based on the UCI Heart Disease dataset, which contains a limited number of samples and clinical features. The use of PCA to reduce the feature space to four components may also affect the information available to the classifier. In addition, the quantum-kernel model is evaluated using simulation, which introduces computational limitations and

does not represent all constraints of physical quantum hardware.

Future research should evaluate the approach using larger and more diverse clinical datasets and compare different quantum feature maps and quantum kernels. Experiments can progressively move from ideal simulation to noisy simulation and, where feasible, physical quantum processors. Further work can also investigate efficient feature encoding, noise mitigation, explainable prediction, hybrid quantum-classical models, and integration with clinical information systems.

IX. CONCLUSION

Quantum Machine Learning provides an emerging approach for investigating disease prediction from structured healthcare data. The reviewed literature shows encouraging results for several QML algorithms, particularly quantum support-vector methods and quantum neural approaches, while also revealing persistent limitations related to hardware, dataset scale, validation, and deployment. In this study, a QML-based heart disease prediction pipeline was implemented using the UCI Heart Disease dataset, combining classical preprocessing and PCA-based feature reduction with a ZZFeatureMap, Fidelity Quantum Kernel, and QSVC. A classical SVC was evaluated using the same matched training subset and test set to provide a controlled comparison.

The study demonstrates the practical implementation and evaluation of a quantum-kernel-based approach for heart disease prediction, while the experimental results should not be interpreted as evidence of a general quantum advantage. Progress toward clinically useful QML will require larger and more diverse datasets, transparent benchmarking, realistic hardware evaluation, efficient quantum representations, and close collaboration between quantum researchers, healthcare professionals, and biomedical experts.

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